

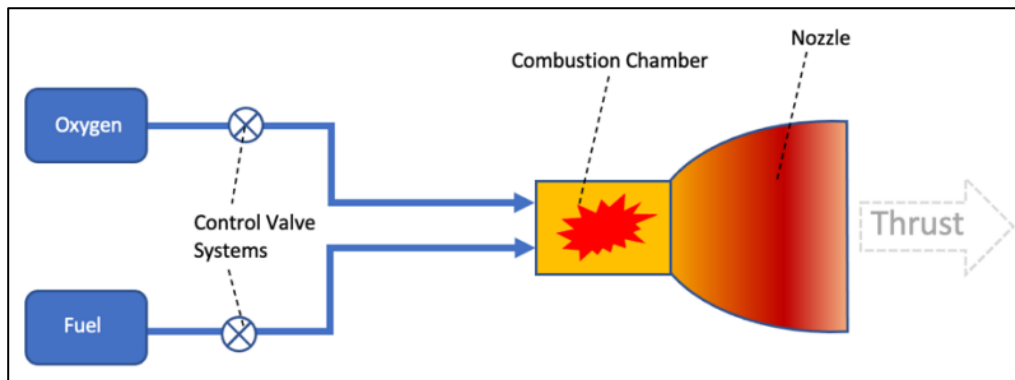


UNIVERSITY
OF WOLLONGONG
AUSTRALIA

LAB REPORT

Subject: MECH470/970

Subsystems and Feedback Control



Week 11

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1. Overview

This lab is an extension of the motor model from Lab 6, incorporating position feedback and closed-loop control. A proportional-integral-derivative (PID) controller is implemented progressively to achieve accurate angular tracking of the valve. A proportional control is introduced, which is then manually tuned using the Ziegler-Nichols method and compared against Simulink's automated PID tuner. An insight into how feedback control improves performance over open-loop operation is highlighted.

2. Results & Discussion

Basic Feedback-based Position Control

Identical to the task in lab 6, the valve assembly is connected to the motor system, and its operation is observed. To simplify the model, the electro-mechanical model is condensed into a subsystem. A basic position-based feedback loop is then created by feeding the motor output back into the input using a 'minus' block. The new model is displayed in *Figure 1*.

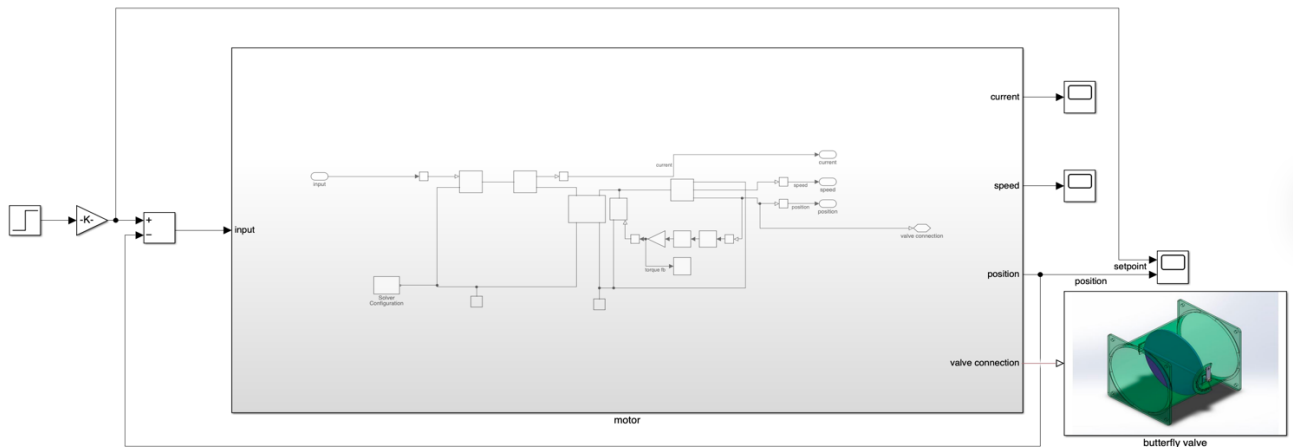


Figure 1 - Basic feedback position control model

To allow the step input to be entered in degrees instead of radians, a gain block is added after the step input with the value of $\pi/180$, allowing the user to enter the value in degrees, where it is mathematically converted back to radians before entering the model. A plot of the system response with basic feedback control is displayed in *Appendix A, Figure 3*.

PID Control and Manual Tuning

To improve the response, a PID control structure is introduced to the error signal as displayed in *Figure 2*.

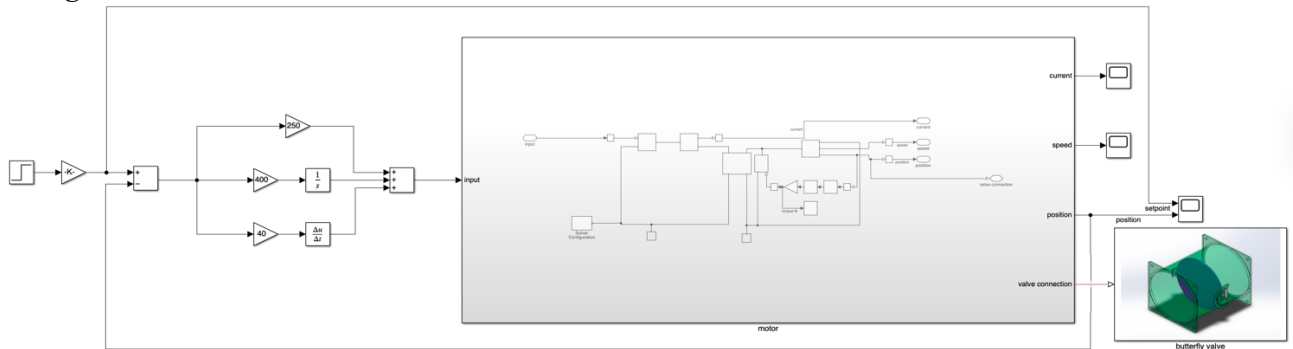


Figure 2 - PID control structure

The Ziegler-Nichols tuning method is used as a baseline to obtain a good response from the implemented PID system. The method derived the following values: $K_p = 166$, $K_u = 277$, $K_d = 10.4$, $T_u = 0.500$ s, $T_d = 0.0625$ s, $K_i = 665$, $T_i = 0.250$ s

The position response of the system with the values determined using the Ziegler-Nichols method is presented in *Appendix A, Figure 4*. Further independent tuning helped achieve the plot displayed in *Appendix A, Figure 5*.

Final Task Trajectory

Since 'P' primarily corrects the signal by producing an output proportional to the instantaneous position error (improving the speed of the corrective response), 'I' eliminates any residual error by accumulating the error over time to ensure the desired position is reached, and 'D' dampens the responding to the rate of change of error, reducing overshoot/oscillatory behaviour - the system response is significantly better as seen by the final task trajectory in *Appendix A, Figure 6*.

Advanced PID Implementation and Tuning

Simulink's PID block with integrated numerical solver is replaced with the existing PID structure to automatically tune the PID parameters for optimal system response. The numerically solved PID parameters are as follows: $P = 13.6$, $I = 4.49$, $D = 9.87$, and N (filter coefficient) = 32.1. While closely aligned with the fully tuned manual response, this system produces the best results with the most aggressive response time as displayed in *Appendix A, Figure 6*.

3. Conclusion & Improvements

Feedback and Suggestions

In addition to the physical modelling by adding mechanical backlash and sensor noise to the model as highlighted in lab 6, a low-pass filter could be applied on the derivative term to suppress any noise amplification. More exploration can also be done on using adaptive PID control to maintain optimal performance across different points. Additionally, feed-forward control may also be added alongside PID feedback, to improve the response time by anticipating changes instead of reacting to them.

Nevertheless, this lab demonstrated the major advantages of the closed-loop control system for precise motor actuation. The different ways of implementing the feedback was explored and the system was able to achieve a significantly faster response with reduced overshoot and near-zero steady-state error when compared to the open loop system.

4. Appendix A

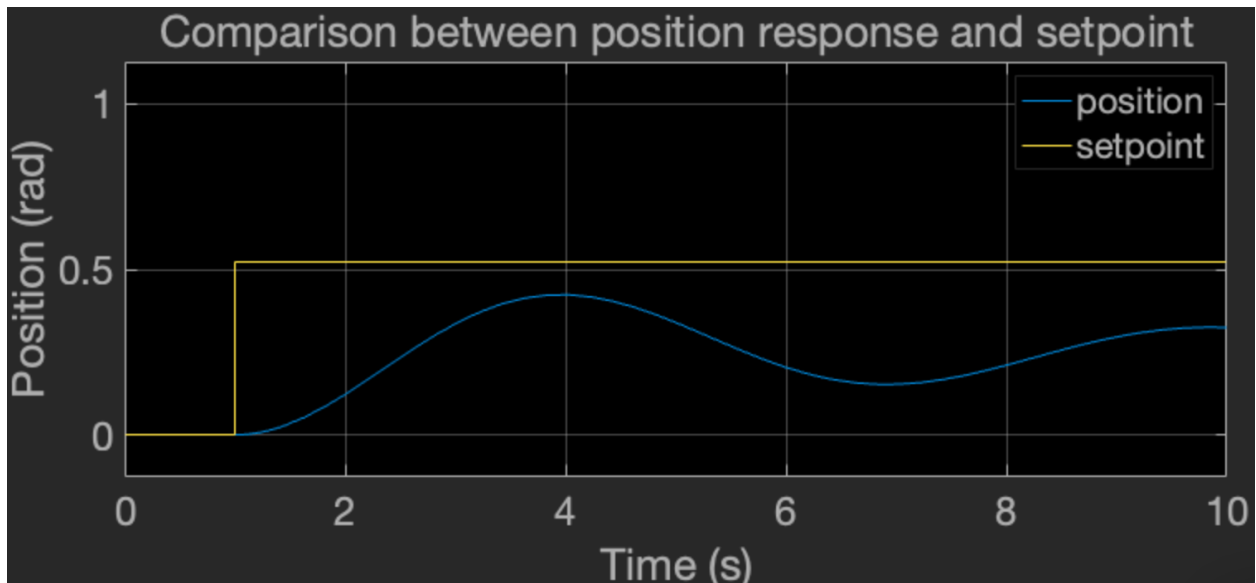


Figure 3 - System response with basic feedback control

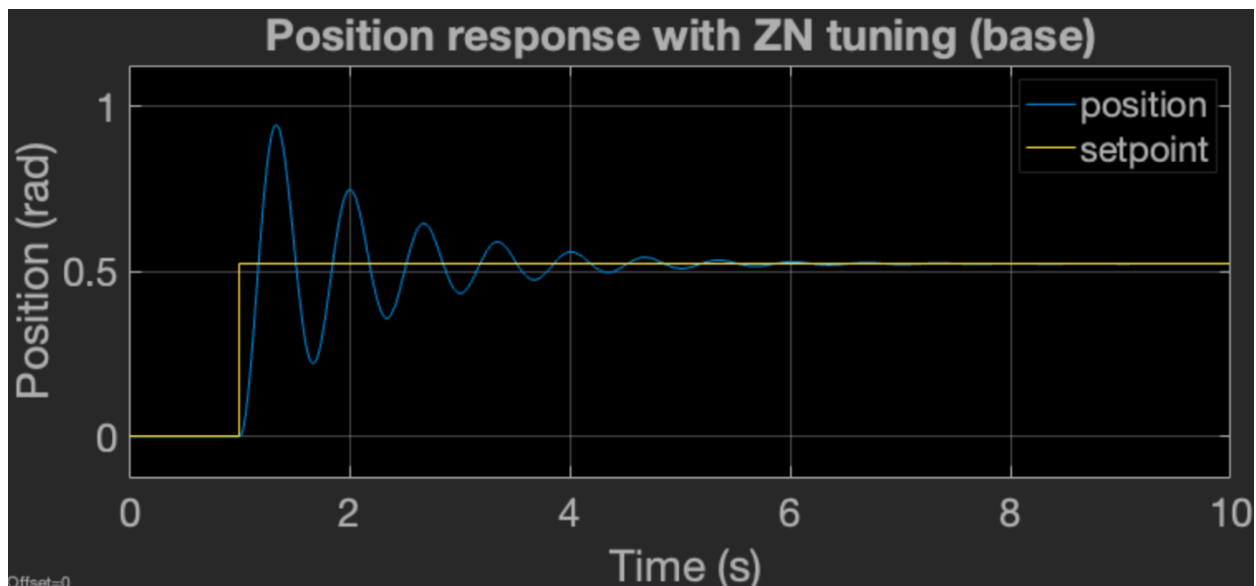


Figure 4 – System response with Ziegler Nichols tuning (baseline)

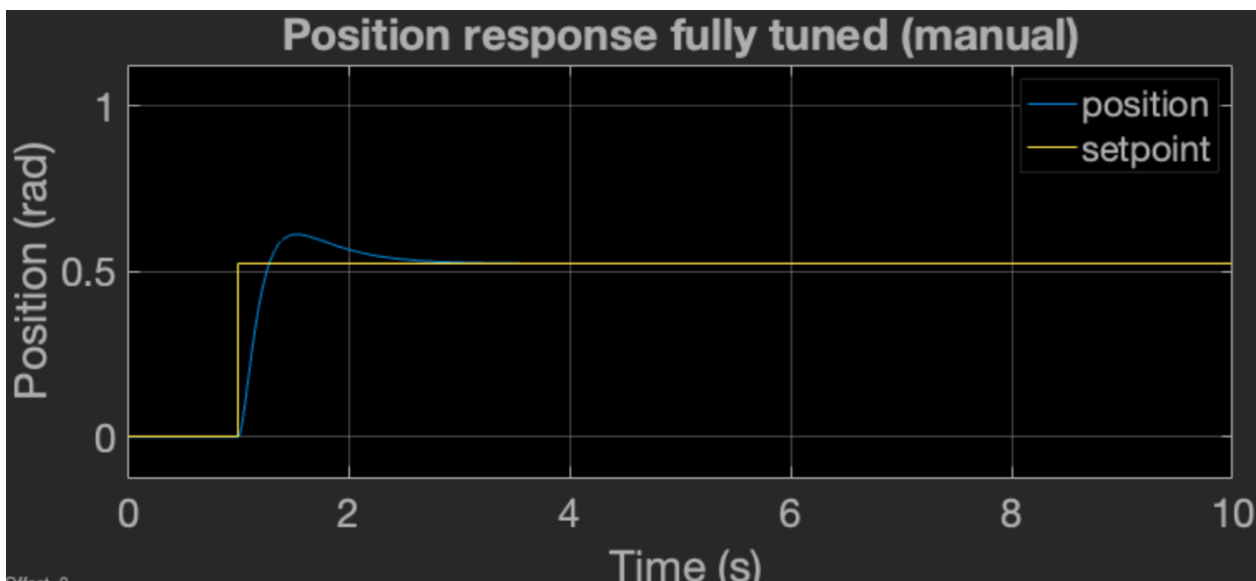


Figure 5 – Fully tuned (manual) system response



Figure 6 – Rocket engine command sequence system response (manually tuned)



Figure 7 - Rocket engine command sequence system response (auto tuned)